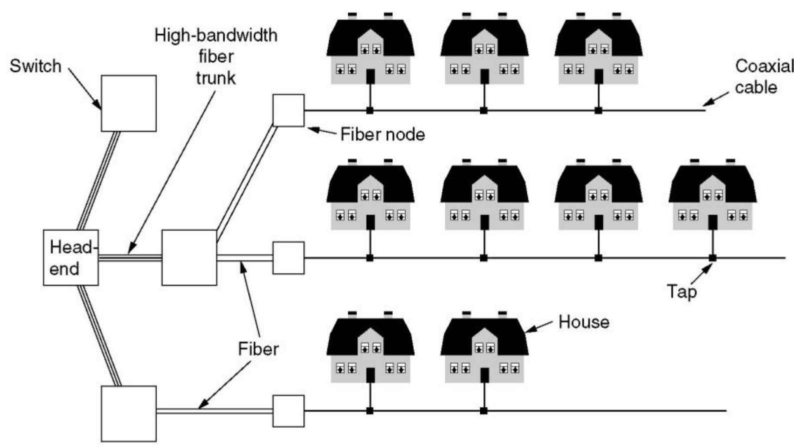
CSE321: Computer Networks
Class Test – 4

Time: 30 minutes **Mark: 20**

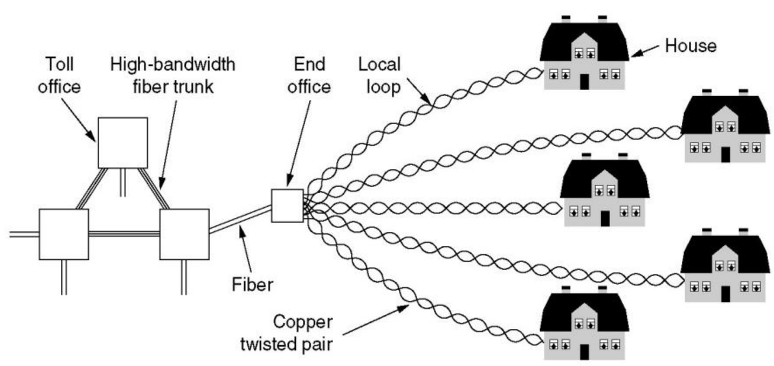
[Instructions:
1) This is an **OPEN-MATERIAL** online examination.
2) Write your answer in a sheet. Make sure you write your Student # there on top of the page(s).
3) Then, make a pdf scanned copy of your answer and submit that (pdf) in Moodle *in time*.
4) Check and confirm at your own that your submitted file is okay and not corrupted.
Grading will be made based on what will be available in Moodle.]

<u>Question:</u>

Different types of topologies are available with fiber trunks as follows.



(a)



(b)

In such cases, the fiber trunks enable high-bandwidth communication. For the purpose of error handling over these high-bandwidth fiber trunks, a mechanism of error correcting code is enabled. In the mechanism, each codeword consists of n bits, where n =m+r. Here, m and r refer to the number of data bots and the number of check bits respectively. The minimum value of r is determined as follows -

$$(m+r+1) \leq 2^{(n-m)}$$

The check bits are positioned in the codeword at the bit indices having power of two (1, 2, 4, and so on). Each check bit is set to ensure odd parity over itself and all other data bits whose indices' Binary representations have the check bit's index as a part. For example, if m=7 and r=4, Check Bit 1 gives odd parity over - (Check bit 1 and Data bit 3, 5, 7, 9, and 11). This happens as 3, 5, 7, 9, and 11 have 1 in their Binary representations (e.g., 3=2+1, 5=4+1, and so on)

Here, k codewords get arranged in a matrix having each codeword along a column and the bits are transmitted along the rows so that error correction can be done even for up to k-bit burst errors.

Now, with the progression of time, the reliability of the trunks get degraded and once it is found that there arises a possibility of happening up to k+1-bit burst errors (instead of previous maximum of k-bit burst errors). Under this circumstance, the error handling criteria is also relaxed. Now, it is needed to make sure only error detection rather than ensuring error correction.

Considering this scenario, you need to answer the following.

1. Is the above method of error correction up to k-bit burst errors sufficient for ensuring error detection up to k+1-bit burst errors?
2. Justify your answer for the above with necessary explanation.